

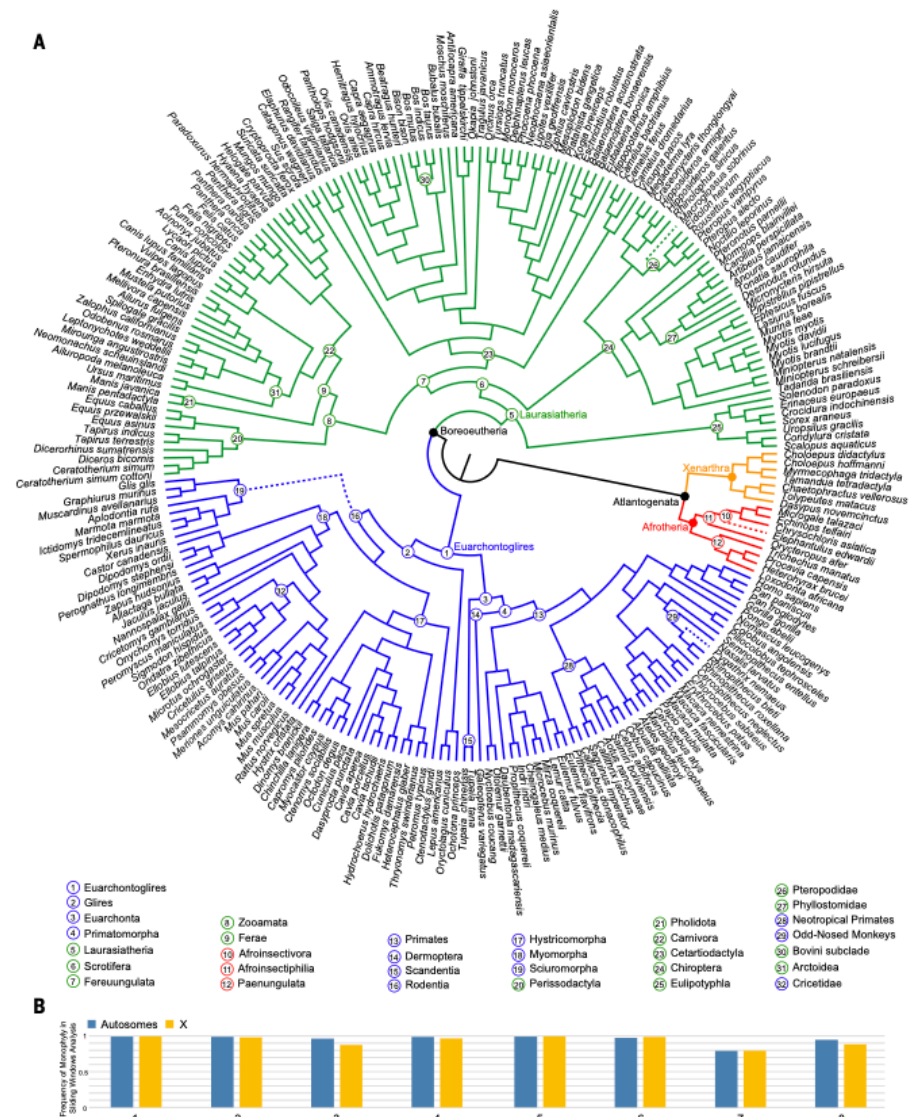
Evolutionary Thinking 2025

Paper discussion of the Zoonomia Paper

Bjarke 090925

Plan

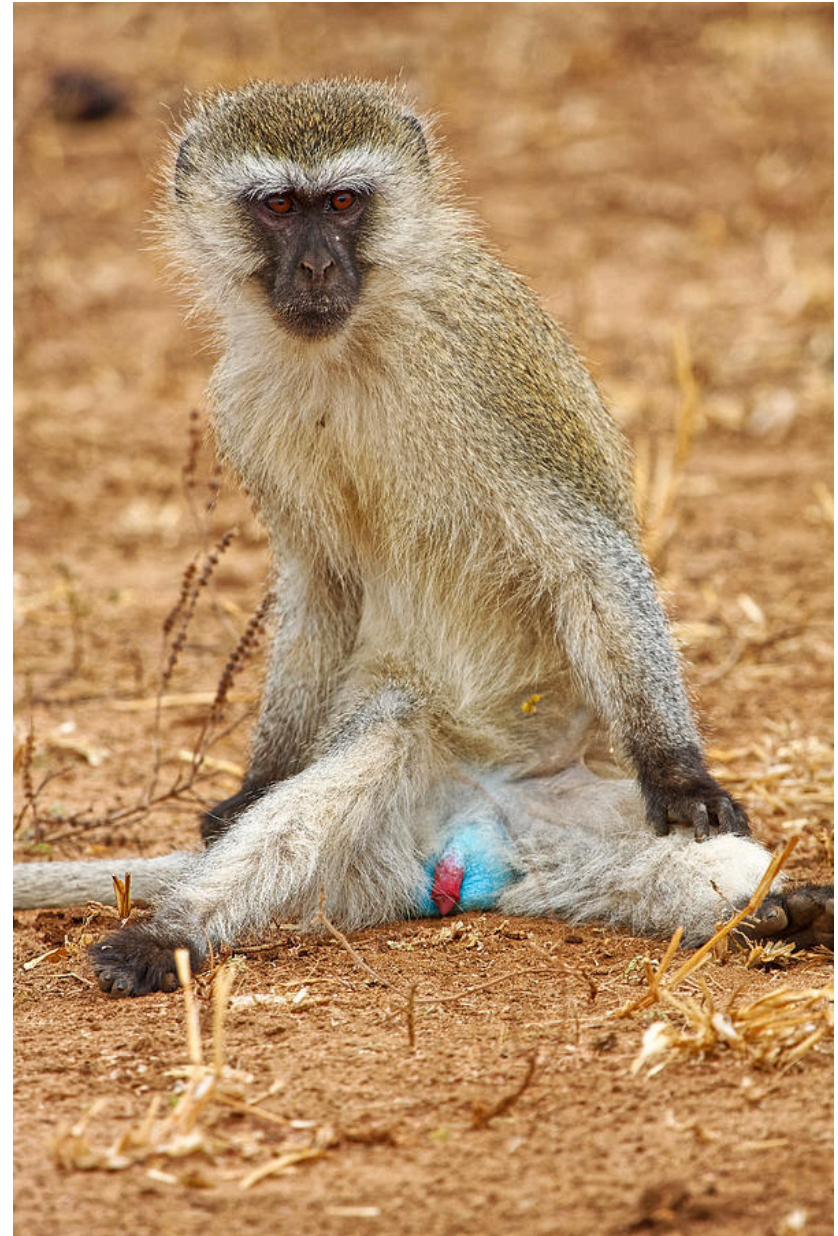
- Primate of the day
- Molecular clocks
- Recalibrate groups :))
- Exercises in groups until 13:00
- Breaking into puzzle 13:15
- Discussion in puzzle
- Recap in class



Vervet monkey

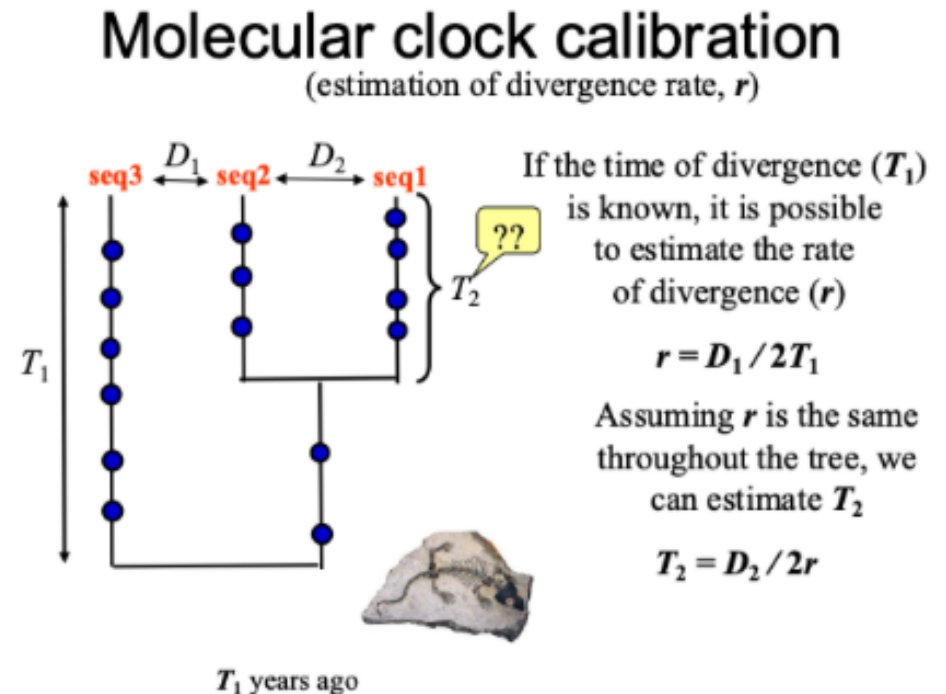
Chlorocebus pygerythrus

Two occasions of raising human children (Robert Mayanja and Ugandan Special Olympics competitor John Ssebunya)



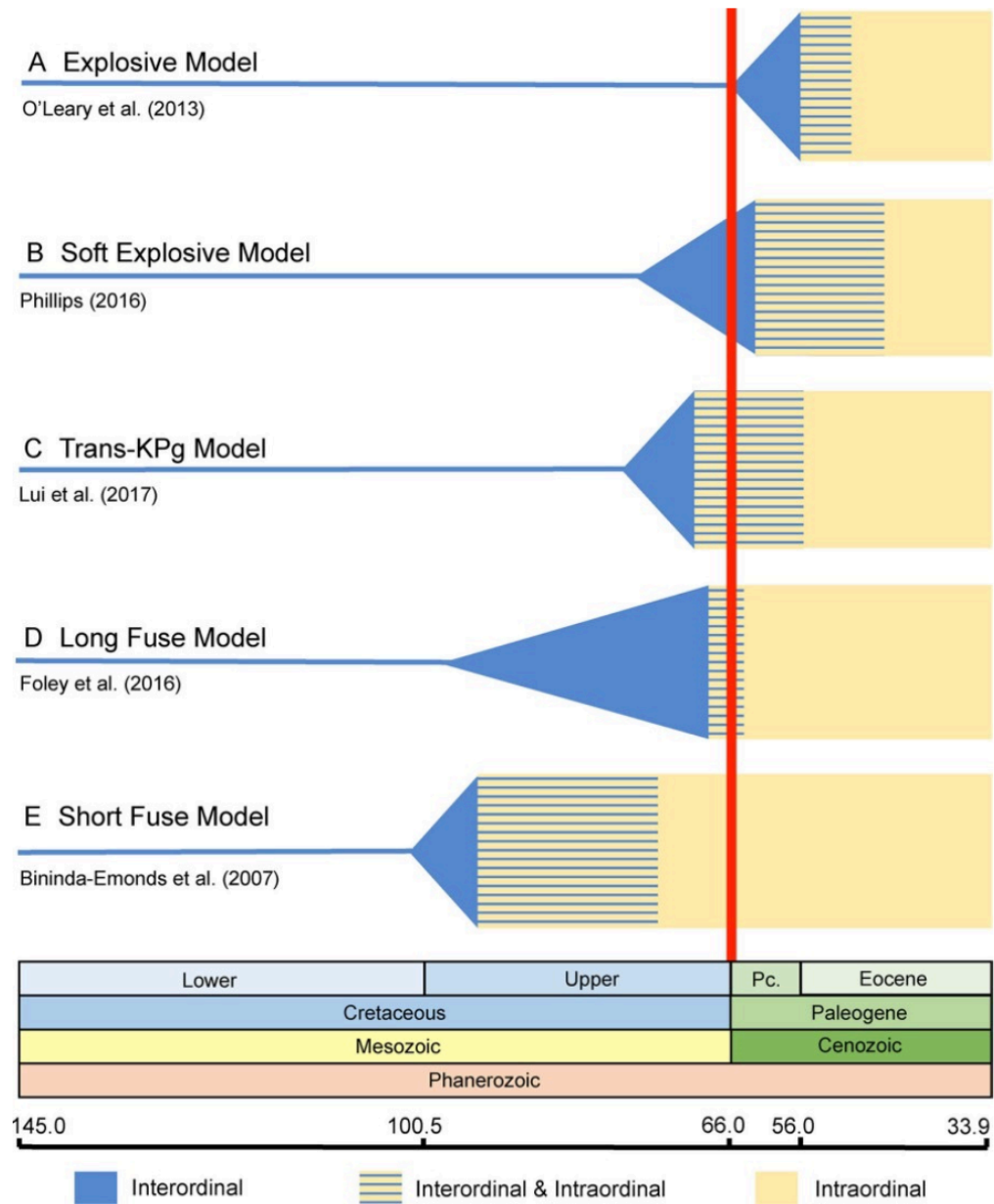
Molecular clocks

- If we think we know the rate of change in sequences, we can get the divergence time between species
- *Only* assumption, rate of change is constant through time and similar between species



**How many are you in each
group?**

Please send me a email with the names of your group so I can update brightspace?



Answer questions

2-3 questions per group

Can't remember the number of groups so we divide them in class

Untill 13:00

A genomic timescale for placental mammal evolution – Questions

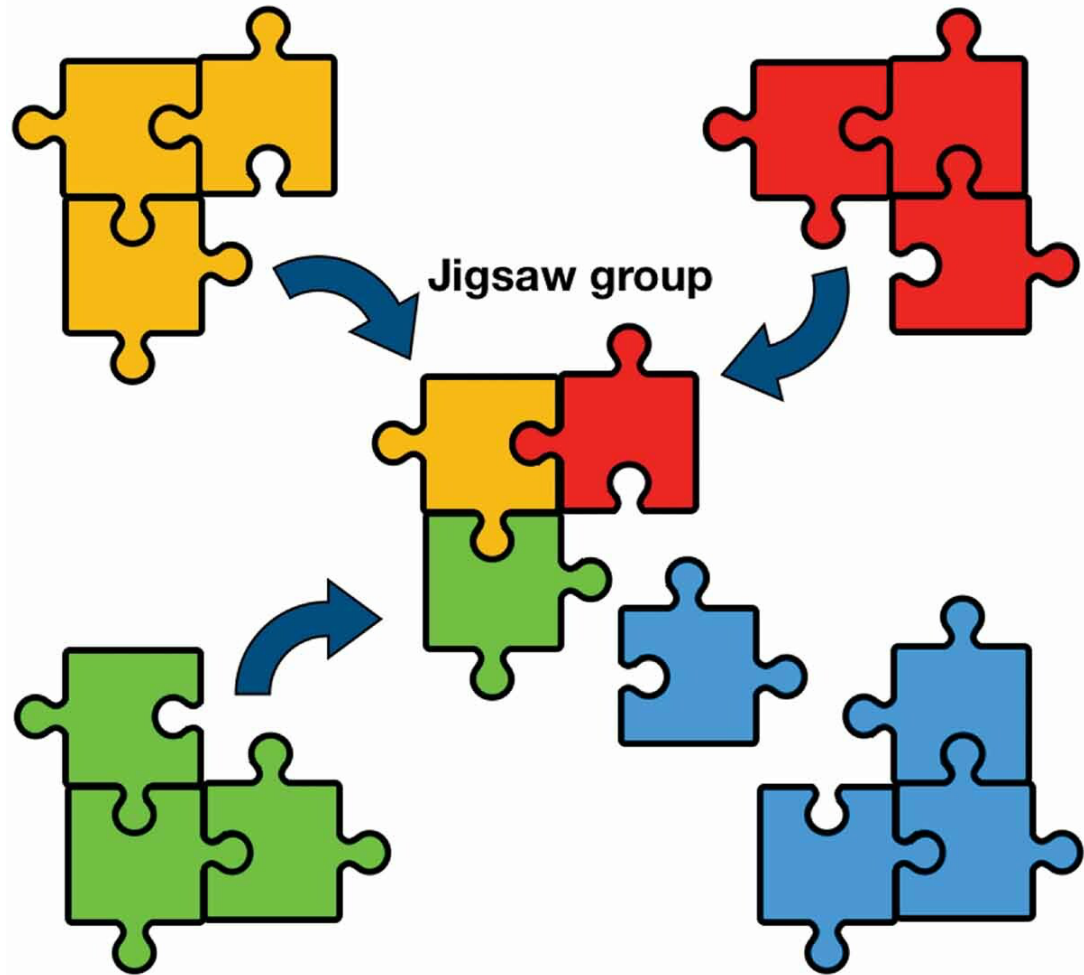
All groups: Questions 1, 2, 3

Groups 1-5: Questions 4-7

Groups 6-9: Questions 8-11

- 1) What are molecular clocks useful for?
- 2) What important assumptions which would influence the robustness of a molecular clock are considered in the paper?
- 3) Why were multiple alignment references employed?
- 4) What important phenomenon potentially leading to conflicting phylogenetic relationships is not directly modeled in the paper? Why is it suspected to contribute to the data analyzed in the paper? What strategy is employed to minimize its impact?
- 5) The authors postulate a prominent difference between superordinal and intraordinal diversification timings and amplitude. How is this motivated?

Expert group



Break till 13:15



Puzzle

